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Journal of Urban Economics 55 (2004) 439–457
JOURNAL OF Urban Economics
www.elsevier.com/locate/jue

Property taxes and property values: evidence from Proposition $2\frac{1}{2}$

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Received 14 February 2003; revised 12 January 2004

Abstract

We examine the effect of Proposition $2\frac{1}{2}$ on revenues and housing prices in Massachusetts. Communities that were initially constrained by the law saw large increases in state aid and the use of fees. We use these initial constraints as instruments for changes in other components of revenue while treating the change in the property tax as exogenously determined by Proposition $2\frac{1}{2}$. Our results strongly suggest that communities that were able to increase their property taxes more rapidly saw greater increases in their housing values in the period following passage of the law.
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1. Introduction

In this paper, we examine the effect of property tax and fee increases on property values in Massachusetts in the aftermath of Proposition $2\frac{1}{2}$, the state's tax limitation measure. We find that communities in which property taxes increased relatively rapidly also experienced relatively rapid increases in residential property values. This suggests that Proposition $2\frac{1}{2}$ constrained taxes to levels below the optimum at least from the perspective of individual communities.

Our approach can be viewed as using the exogenous variation in property taxes induced by Proposition $2\frac{1}{2}$ to test the property-value maximization hypothesis. Under this hypothesis, any law that constrains government from providing its chosen level of public service provision will lower property values in the community.

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doi:10.1016/j.jue.2004.01.002

Of course, property value maximization is not a necessary condition for our findings. If local public goods provision were below the level that maximizes property values, further constraints would also lower property values. Alternatively, the objective function might be more complex or government might face constraints that prevent it from being fully efficient. Thus government might maximize land values subject to providing the mayor or workers some fixed level of rent. Still our results strongly suggest that Proposition $2\frac{1}{2}$ hurt communities that it constrained.

In the next section, we briefly review the major features of Proposition $2\frac{1}{2}$ and its functioning during the period we study. We then provide a brief survey of related literature on property value maximization and the closely related inter-jurisdictional capitalization hypothesis. We argue that Proposition $2\frac{1}{2}$ provides a unique opportunity to address deficiencies in the empirical literature. Sections four and five present the model and data while section six gives the results and section seven concludes.

2. An overview of Proposition $2\frac{1}{2}$

Massachusetts voters passed Proposition $2\frac{1}{2}$ in November 1980. Subject to certain exceptions, the law limited the property tax rate to $2\frac{1}{2}$% of the assessed value of property. Communities with tax rates in excess of $2\frac{1}{2}$% were required to reduce their tax revenues by 15% per year until they complied with the law. Almost half of Massachusetts communities had to cut taxes in the first year the law was in effect. Two communities required four years to reach compliance with the tax rate limitation.

Once the tax rate limit has been met, Proposition $2\frac{1}{2}$ invokes a tax levy limit. Communities can set taxes below the levy limit but may not exceed it. The levy limit on existing property increases by $2\frac{1}{2}$% (nominal) per year. However, if a community chooses not to tax at its limit one year, it does not affect the levy limit. Thus if a community that is taxing at its limit chooses not to increase its taxes one year so that it is approximately $2\frac{1}{2}$% below its limit, the following year it may increase taxes by approximately 5%. New growth in a given year can be added to the tax base and taxed at the community's normal tax rate. Thereafter, it becomes part of the tax base and is subject to the $2\frac{1}{2}$% growth limitation.

Communities may also increase their levy limit by submitting a Proposition $2\frac{1}{2}$ override to the electorate. By the end of the period we study, the levy limit rather than the tax rate limit was the constraining factor for those communities constrained by the law. In principle, therefore, communities were not constrained since they had the option to vote further tax increases. In practice, during the period we study (1984–1988), relatively few communities passed overrides while many were at or near their levy limits. Cutler et al. [8] report that between 1984 and 1990 overrides and exclusions added about 0.1% per year to the tax levy (with such votes being more common towards the end of the period than towards the beginning). This suggests that by requiring an override, the levy limit did constrain taxes.

Proposition $2\frac{1}{2}$ significantly changed both the level and composition of local government revenues. Real per capita local government revenues declined from 1981 to 1984 and did not regain their 1981 level until 1988 despite a booming Massachusetts economy over this period. As of 1992, real per capita local revenues were less than 3% above their pre-Proposition $2\frac{1}{2}$ level. Moreover, real per capita property taxes declined and remained

almost 10% below their pre-Proposition $2\frac{1}{2}$ level in 1992. The loss was made up in part by increased state aid which rose dramatically between 1981 and 1988 and then declined sharply. The rest of the loss was made up by increases in fees and charges which increased by close to 80% going from 11 to 18% of local government revenues between 1981 and 1992.

Proposition $2\frac{1}{2}$ interacts with an earlier law, the Classification Act of 1979, which requires that all properties be assessed at 100% of their value at least every three years. In years in which a community does not conduct a reassessment, the state adjusts the assessed values to calculate the equalized value of property for that community. Despite the Classification Law, prior to Proposition $2\frac{1}{2}$ many communities dragged their feet on reassessing property. The tax limits imposed by Proposition $2\frac{1}{2}$ created an additional impetus to revaluation so that between 1982 and 1984 almost all the communities that had not yet done so adopted 100% valuation and all communities had done so by 1986.

In our empirical work, we will focus on the period between 1984 and 1988. We choose 1984 as our starting date on the basis of a number of considerations:

(1) The initial constraints imposed by Proposition $2\frac{1}{2}$ still have explanatory power for changes in the property tax over this period. Cutler et al. [8] estimate that a \$100 increase in the initial constraint imposed by Proposition $2\frac{1}{2}$ lowered tax revenues by an additional \$13 (from \$86 to \$99 per \$100 initial constraint) between 1984 and 1988 and increased state aid by an additional \$50 and total revenues by an additional \$30.
(2) The adoption of 100% assessment by virtually all communities greatly improves the quality of data starting in 1984. There is evidence that prior to 1984 data on equalized value across communities were very inaccurate.
(3) Although the law had its first impact in 1982, it was still unclear whether it would be subject to significant legislative or judicial changes. By 1984, the future of the law was much clearer.

Between 1984 and 1988, the Massachusetts miracle eliminated the tax rate ceiling as a binding constraint for all communities and relaxed or eliminated the constraint imposed by the levy limit. Thus the period we study is one in which the constraints imposed by the law were loosened. The dramatic end of the Massachusetts miracle in 1989 generated renewed disruption and tightening of the constraints. Using a somewhat different approach, Bradbury et al. [2] examine the effects of the tightening of these constraints between 1990 and 1994.

3. Literature review

This paper draws on two closely related strands of literature: the property-value maximization hypothesis and the inter-jurisdictional capitalization hypothesis. We are concerned with whether differences in taxes and services are capitalized into housing (or more precisely, land) prices, and, if so, whether government chooses the level of services and accompanying taxes that maximizes property values.

The literature on the effects of interjurisdictional variation in taxes or services (e.g. Polinsky and Rubinfeld [16], Case and Grant [6]) has tended to work within a framework derived from the Tiebout model. However, it is well known that other models have similar features. Hobson [11] shows that when the population is fully mobile, differences in tax rates across communities (holding services constant) will be capitalized into land rents. Thus while the burden of the average property tax rate may be shifted among factors of production or to renters depending on supply and substitution elasticities, land rents capture property tax differentials. The link between service levels and taxes on the one hand and property values on the other is quite general (Zodrow and Mieszkowski [23]).

Oates's path-breaking paper [14] was the first to confront the inter-jurisdictional capitalization hypothesis empirically. He argued that the Tiebout [20] model implies that services will increase property values while taxes will reduce them and confirmed these predictions. However, Edel and Sclar [9], Hamilton [10] and Meadows [13] pointed out that in Tiebout equilibrium land values are independent of the level of service provided. Therefore, either the New Jersey municipalities were not in equilibrium or the Tiebout model requires modification. The ensuing research has tended to present itself as either examining disequilibrium or studying whether service and tax differentials are capitalized into housing values without directly relating the research to the Tiebout model (see Shenkel and Gaines [18], Brueckner [3], Taylor [19] and related work by Ladd and Bradbury [12]).

Testing for capitalization within a Tiebout model is problematic, because the model suggests no obvious reason why public services and tax levels should not be perfectly related. Under the Tiebout model, either service levels or tax levels is a sufficient statistic for public good provision (Hamilton [10]). Conditional on the service level, the tax rate is therefore purely a measure of house values. Since housing values are in the denominator of the tax rate, the tax rate variable will be negatively correlated with housing values. Since tax levels and service levels are positively correlated, OLS estimates will tend to generate a positive coefficient on service levels and a negative coefficient on tax rates even if we are in full Tiebout equilibrium.

Oates recognized the difficulties with the endogeneity of taxes and services and proposed a 2SLS estimator. However, Pollakowski [15] challenged the choice of instruments. We can translate his argument into somewhat more modern language. There is a hedonic relation between housing prices and housing characteristics including the local public finance characteristics. If we need to instrument for local taxes and services, we must find variables that influence their levels but are not correlated with the error term. Since the error term includes any unmeasured housing characteristics, it is difficult to think of factors that affect the demand for local public goods but for which we can convincingly argue that they do not affect the demand for other housing characteristics. In the absence of the Tiebout model, we face the opposite problem. Without a well-formulated theory of why spending and service levels differ, it is difficult to argue convincingly whether instruments are needed or what are potential instruments.

The research in this paper is closely related to Brueckner [4] which, drawing on Brueckner [5], argues that if public goods provision is efficient, differences in levels of public goods provision should not affect property values. He finds no relation between expenditure and property values for a cross-section of Massachusetts communities in

1976. That paper relies on an instrumental variables procedure to avoid the possibility that unmeasured characteristics raise both the demand for public goods and property values.

The approach taken here is somewhat different. We use the experiment induced by Proposition $2\frac{1}{2}$ to address concerns about the existence of suitable instruments. We treat Proposition $2\frac{1}{2}$ as a constraint on government optimization. Essentially Proposition $2\frac{1}{2}$ determines property tax changes for many jurisdiction. If governments seek to maximize property values in their constituencies, then more heavily constrained communities will be further from the optimum and hence will suffer larger declines in housing prices. The approach used here is closest to Rosen [17] who examines the capitalization of tax rate reductions following the passage of Proposition 13 in California. Rosen argues that the state bailout left the change in services nearly uniform across communities while impacting tax rates differently.

4. The model

The model developed in this section is oriented towards laying the ground work for our empirical analysis. It therefore does not include much of the richness of recent research on Tiebout economies (see Conley and Wooders [7] for a review). In particular, since we consider the effects of Proposition $2\frac{1}{2}$ over a short period, we treat both the aggregate housing stock and its distribution across communities as fixed. We also ignore consumer heterogeneity and concentrate on the consequences of heterogeneity across communities.

In the context of this simple model, we show that if, in the absence of constraints, communities provide the optimal level of public goods, then communities that are more heavily constrained will experience larger drops in their property values. While we derive a justification for estimating an equation in differences, the approach is closely related to Brueckner [4,5] who shows the equivalence between property-value maximization and efficiency of public good provision and uses this relation to test efficiency by examining the relation between spending and property values in a cross-section.

Each community provides a public good which we think of as education since most Massachusetts communities spend over half their budget on education. Public education must be provided to all members of the community on an equal basis. However, the cost of providing education of a given quality is a fixed amount per member of the population. Thus the public good is really a publicly provided private good or in Bewley's [1] terminology, a pure public service. Since the size of each community is fixed, the community cares only about the average (or total) cost of providing education to its residents. We thus ignore aggregate crowding issues. The key assumptions, A1–A5, are described below.

A 1. The cost of producing the public good in community i is cG per member of the community where c is the marginal cost of producing the good and G is the number of units of the public good provided to each member of the community.

An earlier version of this paper allowed c to vary across communities. In the absence of cost heterogeneity, since individuals are identical, communities will all choose the same

level of public goods in the absence of constraints. This is obviously unrealistic but all of the essential points can be made without adding the complexity of heterogeneity to the model.

A 2. The public good may be financed through property taxes, T, state aid, A, and/or fees, F. Property taxes are subsidized at rate t through the income tax system while fees are not so that the net cost to residents is $(1-t)T+F$. The full rental cost of housing is $h+(1-t)T+F$. Since houses are identical in the model, there is no loss from treating costs of upkeep as zero. Then h is the implicit rental cost of housing over and above payments for taxes and fees. Since property value is just the discounted value of the stream of net returns, h is the flow of value from property or, equivalently in equilibrium, foregone income from owning and occupying the property.

It is evident that fees are dominated by property taxes in the absence of constraints on property taxation. Fees will only be used in the presence of a measure limiting the tax rate and therefore will be ignored for the moment.

A 3. In addition to the public good, there is a homogeneous private good that serves as numeraire. There is a large number of homogeneous individuals with differentiable utility $u(Y-p,G)$ where Y is exogenous income and p is the rental cost of housing including taxes. u is assumed to be quasi-concave and twice differentiable with $(\partial u/\partial Y)_{Y-p=0} = (\partial u/\partial G)_{G=0} = \infty$.

It should be noted that Y is assumed to be independent of the level of the public good. Massachusetts communities are small relative to the labor market in which they operate. We therefore do not believe that the provision of public goods to residents in any one community could affect wages either in that community relative to other communities or in the labor market as a whole. In principle, providing public goods to workers in a community could affect wages, but the goods provided by Massachusetts communities are overwhelmingly oriented towards residents. In particular, spending on education typically accounts for half or more of all local spending.

A 4. The housing stock is fixed and homogeneous.

Thus, we abstract from the effects of property taxation on the quality of housing. Since we will be examining interjurisdictional effects within Massachusetts, it seems safe to ignore the impact of the law on the return to capital in the United States. On the other hand, it is possible that Proposition $2\frac{1}{2}$ affected the nature of housing in the state. Our empirical work covers the period from 1984 to 1988. The stock of housing was large relative to new construction over this period. If anything the reduction in taxes should have increased new construction. We therefore believe that we will not violate reality significantly if we treat the housing stock as fixed.

A 5. Each community chooses the level of the public good to maximize the utility of a representative resident.

No individual will want to change communities if

$$u\big(Y-(1-t)T-F-h,G\big)=u^* \tag{1}$$

or, imposing the government budget constraint,

$$u^*=u\big(Y-h-(1-t)T-F,[A+T+F]/c\big). \tag{2}$$

The assumption that individuals are homogeneous is clearly strong, but it largely serves to eliminate technical difficulties. Under a condition such as that used by Westhoff [22] to ensure sorting of types in equilibrium, Eq. (1) will hold for marginal individuals.

To see the relation between housing prices and the different elements of municipal financing, fully differentiate (2) to get

$$0=-u_1\,\mathrm{d}h+\left(\frac{u_2}{c}-(1-t)u_1\right)\mathrm{d}T+\left(\frac{u_2}{c}-u_1\right)\mathrm{d}F+\frac{u_2}{c}\,\mathrm{d}A. \tag{3}$$

The differential equation (3) does not determine the overall housing price level, only differences across communities. To complete the determination of housing prices, we require an outside option such as the ability to build in an empty community. Since we will be differencing out the overall housing price level in our empirical work, we do not formally model an outside option. For a brief discussion, see our earlier working paper version of this article (available on Lang's website).

Recall that h is the flow value of housing services. Owner-occupiers will want to maximize h which is equivalent to maximizing the utility of the residents of the community holding h constant.[1] Maximizing u with respect to T and F gives the conditions

$$(1-t)cu_1=u_2, \tag{4}$$

$$F=0. \tag{5}$$

If communities are heterogeneous but sufficiently homogeneous, the choice of the public good in each community will satisfy (4) approximately for each resident.

It should be apparent that all communities will choose the same level of T and F and thus the same housing value if they have the same c and A. The working paper version of this article examines the case where c varies across communities. However, since we will be focusing on the effect of changes in A, T, and F, this complexity is unnecessary.

Examining (3) and (4) confirms that if government sets taxes to maximize utility, then any departure from its optimal choice will lower property values (h). If government is constrained to set taxes too low, then $u_2/c-(1-t)u_1>0$, so that, from (3), h is increasing in T, or equivalently relaxing the constraint will increase property values. Conversely, if government sets taxes too high, raising taxes will lower property values. In addition, when taxes are set optimally $u_2/c-u_1=-tu_1<0$ and from (3), raising fees lowers property values. However, if taxes are constrained sufficiently that $u_2/c-u_1>0$ when F equals 0,

[1] If we had renters in the model, they also would want to maximize h provided that their rent was fixed over some term. If rents changed instantaneously and the number of communities were infinite, renters would be indifferent to the choice of T and F. With a finite number of communities, improving the efficiency of public good provision in any one community would raise u^* and thus make all renters better off. Hence conditions (4) and (5) are quite general.

then raising fees will increase h until they reach the level that $u_2/c = u_1$. If government is efficient in its choice of fees, it will either set them to 0 (the outcome when taxes are not too heavily constrained) or raise them to the point where their marginal effect on property values is 0 (the outcome when taxes are heavily constrained). Not surprisingly, increasing government aid raises these values.

Under Proposition $2\frac{1}{2}$ communities are constrained to keep their tax rate below 0.025 so that $T < 0.025h$. If this constraint is binding, we have

$$u_2 > u_1(1-t)c. \tag{6}$$

Optimizing with respect to F yields

$$u_2 \leqslant u_1 c \tag{7}$$

with strict equality if $F > 0$.

It is not clear to what extent communities in Massachusetts are actually free to set fees optimally. State law prevents fees from exceeding the cost of providing the service. There is some flexibility in defining the service (e.g. recreation versus golf, tennis, basketball, etc.) and some ability to inflate the cost of services by decisions regarding the allocation of fixed costs. Still, we are aware that at least some communities viewed themselves as constrained by the limit on fees. If communities are constrained from setting fees optimally given the constraint on taxes, then increasing fees will also raise property values. If communities are not constrained, then at the optimum, there is no effect of fees on property values.

Finally, although we have modeled fees as independent of use of the public good, this is not always the case. Some depend on use as in the case where there is a daily fee for use of the local swimming pool. Others, provide access to a public good but do not depend on use. Thus, for example, many Massachusetts communities have a flat annual fee for trash removal. As we would expect from the model, local receipts (mostly fees and fines) rose significantly following the passage of Proposition $2\frac{1}{2}$. Our discussions with local officials suggest that the largest growth came from establishing trash fees which closely approximates the increase in fees modeled above and from very large increases in water and sewage fees related to the Boston Harbor clean-up, which does not. If demand for services financed through services is elastic and if the marginal cost of providing the service is positive, fees set somewhat below marginal cost (to take account of tax advantages) will be efficient.

5. Data and estimation

The data are drawn from the Massachusetts Department of Revenue (DOR) municipal data bank. The data bank includes information on tax rates, revenues by source (e.g. property taxes, fees and fines, state aid), equalized value of property and the date of the assessment on which property values are based. Massachusetts communities are required to reassess all property at 100% of value at least every three years. The DOR releases equalized value estimates for each community every two years, adjusting the most recent assessment for developments in the property market since that assessment took place.

We measure property values by the DOR's estimate of the equalized value of property divided by 1990 population estimates from the US Census. This measure has the advantage of being available for the period prior to Proposition $2\frac{1}{2}$, thereby allowing us to perform certain checks on our estimation. It is also available for all the observations in our sample. In contrast, the assessed value of residential housing was not available for 20% of our sample. Moreover, individuals at the Department of Revenue expressed concern about the accuracy of the assessed value of residential housing data in 1984. Nevertheless, since we do not have a good measure of nonresidential development, there is a risk that this measure will capture the growth of commercial property rather than an increase in the value of existing property. Across Massachusetts residential property accounts for over three-quarters of total assessed value. Nevertheless, if there were significant areas of commercial development not accompanied by new housing, this could cause a problem for our estimates. We address this issue below.

By law, assessments must be based on sales of property occurring within a narrow window around the date of the assessment. As a consequence, communities with few transactions have inaccurate assessments. These are reflected in large changes in the average assessment. Communities with fewer than forty property transactions in 1984 are excluded from our restricted sample. This results in a loss of 76 small communities from the sample all with populations below 6100. We further exclude from our restricted sample thirteen communities that had not complied with the Classification Law by 1984.

The model applies to communities with a fixed housing stock and to the change in the value of a fixed stock. The results below exclude communities in which the number of housing units increased by more than 20% between the 1980 and 1990 censuses. This results in the loss of 84 communities ranging in population from 1481 to 49,832 from the sample. The restricted sample contains 178 communities. While we focus on this restricted sample, we also present results based on all 351 communities.

Our basic strategy is to estimate an empirical version of Eq. (3) in which the change in equalized value is the dependent variable and changes in the components of revenue are the principal explanatory variables:

$$\begin{aligned}\%\Delta(\text{eqv}) = a + b * \Delta(\text{Real levy per capita}) + c * \Delta(\text{Real receipts per capita})\\ + e * \Delta(\text{Real aid per capita}) + \text{community type dummies}\\ + \text{assessment year dummies.}\end{aligned} \tag{8}$$

Unfortunately, due to data limitations, we must use receipts as a proxy for fees. Local receipts include fines and motor vehicle excise tax receipts (set at 2.5% by law) as well as fees.

The final issue we must address is endogeneity of the revenue variables. Since the state aid formula explicitly relies on equalized value as one determinant, the change in state aid is clearly endogenous. Formal tests confirm this.

Proposition $2\frac{1}{2}$ largely eliminated the endogeneity of property tax revenues by initially forcing some communities to reduce their tax rates and then by constraining the rate of growth of tax revenues. More importantly, in contrast to the practice in many areas, it decouples tax revenue from property values. When property values rise, the tax rate must be reduced to keep tax revenues constant except for the permitted 2.5% growth rate. While

not all communities were at their levy limit, over the period we study housing prices were rising sufficiently rapidly that almost all communities would have exceeded their levy limit if they did not adjust the tax rate. Thus there is no automatic relation between property values and tax revenues.

There are three reasons that the tax levy may nevertheless still be endogenous. First, communities are not required to collect all the tax revenues permitted by Proposition $2\frac{1}{2}$, and some choose not to. Secondly, the limit is adjusted for new development. New development will therefore simultaneously raise tax revenues and equalized value. We experimented with using open space (unimproved land) in 1984 as a proxy for development potential and including it in the equation. However, preliminary estimates found no relation between open space and equalized value growth, and we dropped this direction of inquiry. Finally, a community is allowed to exceed the restriction on the growth of tax revenues (but not the restriction on the tax rate) by a vote of 50% of the participating electorate, but as discussed in the section on Proposition $2\frac{1}{2}$, in its early years such overrides were infrequent and less common when property values were falling. Thus failing to control for the endogeneity of overrides may create a small downward bias in the estimated impact of taxes on property values.

Similarly, local receipts may be endogenous if they respond to changes in property values. If people buy more or less expensive cars when the cost of property increases, there may be an endogeneity problem.

We have a number of potential instruments. Between 1984 and 1988, state aid increased rapidly and was aimed at those communities impacted most severely by Proposition $2\frac{1}{2}$. We therefore include measures of whether the community was constrained to cut taxes in 1982 and 1983 as potential instruments. The severity of the cuts required by Proposition $2\frac{1}{2}$ would be less in a community which could lower its tax rate by moving to 100% assessment. We estimate the "true tax rate" as the tax levy in 1981 divided by mean of the 1980 and 1982 equalized values. The true rate and its square are potential instruments. Finally, the existence of open space may have allowed communities to avoid the limits of Proposition $2\frac{1}{2}$ by tailoring development to their needs. We experiment with the ratio of the value of open space to other property values as an instrument. Our strategy is to treat the change in state aid as endogenous and then test for the endogeneity of the two remaining revenue variables.

Descriptive statistics for key variables are given in Table 1. It is apparent that spending per capita rose rapidly during this period even adjusted for inflation. This reflects the recovery from the 1982 recession and the booming economy of the Massachusetts miracle. Percentage increases in state aid and fees were much more rapid than increases in the property tax levy.

Before turning to the results, it is important to note that they can only tell us about the impact of changes in relative revenues among Massachusetts communities. Because all of the variation in the data occurs within Massachusetts, we cannot address the impact of Proposition $2\frac{1}{2}$ on the state as a whole. It is possible that the law affected all communities positively or negatively (for example by increasing or reducing the demand for labor across the state). Our work cannot address that issue directly.

Table 1
Descriptive statistics

	Mean	Max	Min	Std. dev.
Constrained, 1982	0.54	1.00	0.00	0.50
Constrained, 1983	0.13	1.00	0.00	0.34
Per capita aid, 1984	222.32	552.62	40.36	87.39
Per capita aid, 1988	337.64	842.18	61.33	144.22
Per capita levy, 1984	506.59	1145.36	135.40	196.90
Per capita levy, 1988	632.66	1518.50	181.28	249.05
Per capita receipts, 1984	112.16	1063.75	24.82	117.27
Per capita receipts, 1988	178.23	1332.07	51.94	151.94
Equalized value per capita, 1984	26366.32	87151.71	10568.85	12452.36
Equalized value per capita, 1988	64153.53	191231.10	21392.63	29662.26
Certification year				
1982	0.34	1.00	0.00	0.47
1983	0.50	1.00	0.00	0.50
1984	0.33	1.00	0.00	0.47
1986	0.44	1.00	0.00	0.50
1987	0.37	1.00	0.00	0.48
1988	0.20	1.00	0.00	0.40

6. Results

6.1. Base specification

Table 2 gives the results of the base model. The top panel provides estimates for the restricted sample while the lower panel uses all 351 communities. Column (1) provides the naive estimates in which all revenue variables are presumed to be exogenous. The results indicate that tax levy increases raise property values while state aid increases do not and the effect of fee (local receipt) increases differ between the two samples. The surprising absence of a state aid effect is readily explained by the endogeneity of state aid which biases the OLS coefficient. Looking at column (2) which instruments for state aid,[2] it is apparent that the OLS specification is easily rejected. When the change in aid is treated as endogenous, increases in state aid have a large positive effect on property values. The results for the effect of taxes and fees are essentially unchanged.

We view the sparseness of the specification as an advantage because it avoids concerns about the possible endogeneity of community characteristics to housing prices. Still column (3) adds community characteristics (see footnote to the table) to the specification in column (2). The substance of the results is again unchanged although the effect of aid is measured with considerably less precision in the full sample and the effect of changes in fees is diminished in both samples and turns noticeably, albeit insignificantly, negative in the full sample.

[2] We considered five potential instruments: constrained in 1982, constrained in 1983, the true tax rate in 1981, and its square and open space in 1984. Preliminary estimates established that only the two constraint variables had explanatory power in the change in state aid equation. The results are similar when all five instruments are used.

Table 2
Determinants of percentage increase in equalized value (1984–1988)

	(1)	(2)	(3)	(4)
		Restricted sample ($N = 178$)		
ΔLevy/100	0.22	0.27	0.23	0.02
	(0.05)	(0.07)	(0.07)	(0.20)
ΔAid/100	0.07	0.53	0.59	0.44
	(0.05)	(0.20)	(0.23)	(0.20)
ΔReceipts/100	0.19	0.22	0.14	0.51
	(0.07)	(0.09)	(0.09)	(0.47)
F-test on excluded instruments				
Aid	–	8.78	6.10	8.78
Levy	–	–	–	3.81
Receipts	–	–	–	1.22
OID test	20.8	0.25	0.91	5.49
		Full sample		
ΔLevy/100	0.11	0.12	0.10	−0.06
	(0.05)	(0.02)	(0.02)	(0.07)
ΔAid/100	0.09	0.52	0.84	0.46
	(0.05)	(0.20)	(0.35)	(0.27)
ΔReceipts/100	0.08	−0.00	−0.14	−0.22
	(0.05)	(0.07)	(0.11)	(0.27)
F-test on excluded instruments				
Aid	–	11.07	5.51	4.54
Levy	–	–	–	14.52
Receipts	–	–	–	3.92
OID test	38.65	0.63	0.11	5.46
N	351	351	351	335

Notes. Two-stage least squares estimates. The explanatory variables are expressed in real per capita terms. All specifications include six community type dummies and six dummies for the timing of the assessments. Standard errors are in parentheses. (1) None endogenous, additional first-stage regressors for purposes of OID test—constrained in 1982, constrained in 1983, true tax rate in 1981, and its square, open space ratio in 1984; (2) aid endogenous, additional first-stage regressors—constrained in 1982, constrained in 1983; (3) same specification as (2) with median year housing built, %high school graduates, %college graduates, %executives and professionals, %white collar, median family income, and per capita income, as additional exogenous explanatory variables; (4) aid, levy, receipts endogenous, additional first-stage regressors—constrained in 1982, constrained in 1983, true tax rate in 1981, and its square, open space ratio in 1984.

In neither case do we find any evidence that the tax levy or local receipts should be treated as endogenous. In both cases, the Basmann–Sargan test of the overidentifying restrictions is below its expected value and well below conventional levels of significance. Nevertheless, column (4) attempts to treat both variables as endogenous. The main substantive result from column (4) is that when we treat all sources of revenue as endogenous, the data are insufficient to provide us with any real insight. The estimates are very imprecise and the model does not appear to be identified. Moreover, the Basmann–Sargan OID test-statistic is significant at the 0.1 level, suggesting that one of the additional instruments used to identify the model when both revenue variables are endogenous is, in fact, correlated with the error term. We use the Durbin–Wu–Hausman test to determine whether the estimates differ between columns (2) and column (4). The test statistic falls

well short of significance using the restricted sample but indicates that the estimates are different using the full sample. Visually, we can see that in the full sample, the coefficient on the tax levy changes significantly. While we are inclined to rely on the OID test to conclude that the problem lies in inappropriate instruments rather than the endogeneity of the levy, the results suggest caution when analyzing the results based on the full sample, leading us to focus on the results for the restricted sample.

6.2. Constrained communities

As a check, we further restrict the sample to ensure that we capture only exogenous changes in the tax levy. In Table 3, we rely on the degree of excess taxing capacity to further restrict the sample. In principle, a constrained community should be at 100% of its levy limit. However, practical factors may prevent the community from reaching this theoretical maximum. We treat communities as constrained if they were within 1% (columns (1)–(2)) or 0.1% (columns (3)–(4)) of their levy limit in either 1985 or 1988. Unfortunately, data on excess taxing capacity do not appear to be available prior to 1985.

If anything, the results indicate larger positive effects of tax increases as we limit the sample to communities that were constrained by the law at either the beginning or end of the period. For the restricted sample, the coefficient on the tax levy goes from 0.27 in column (2) of Table 2 to 0.35 in column (1) of Table 3 and to 0.37 in column (3). Similarly, using the full sample, the coefficient goes from 0.12 in Table 2 and column (2) of Table 3 to 0.20 in the last column. We note in passing that with these restrictions the effect of increases in fees is never statistically significant but is also imprecisely measured.

6.3. Residential property values

So far, we have used changes in equalized value as our measure of property values. We have done so because we are concerned that assessed values may be unreliable. Prior to

Table 3
Determinants of percentage increase in equalized value (1984–1988)

	(1)	(2)	(3)	(4)
ΔLevy/100	0.35	0.12	0.37	0.20
	(0.12)	(0.04)	(0.12)	(0.05)
ΔAid/100	0.85	0.60	0.68	0.47
	(0.36)	(0.30)	(0.36)	(0.29)
ΔReceipts/100	0.21	−0.09	0.14	−0.15
	(0.13)	(0.12)	(0.15)	(0.14)
F-test excluded inst.	4.88	6.14	3.41	5.29
OID test	0.50	0.33	0.29	0.42
N	121	219	83	140

Notes. Two-stage least squares estimates. The explanatory variables are expressed in real per capita terms. All specifications include six community type dummies and six dummies for the timing of the assessments. State aid is treated as endogenous. Constrained by Proposition 2½ in 1982 and 1983 are additional first-stage regressors. Standard errors are in parentheses. (1) Within 1% of levy limit in either 1985 or 1988, restricted sample; (2) as in (1) but otherwise full sample; (3) within 0.1% of levy limit in either 1985 or 1988, restricted sample; (4) as in (3) but otherwise full sample.

Proposition $2\frac{1}{2}$, communities had an incentive to undervalue property. After 1979, they were prohibited by law from doing so, and by 1984 most communities were in apparent conformity with the law. After Proposition $2\frac{1}{2}$, communities that were constrained by the $2\frac{1}{2}$% tax limit had an incentive to over-assess property. Thus we could easily find that those communities that reduced property taxes the most, also increased assessments the most even if there were no real impact on property values. Nevertheless, relying on equalized values has the disadvantage that it includes the value of commercial and industrial property and open space. While we have restricted our sample to communities with little residential growth, we cannot preclude the possibility that changes in equalized value and the tax levy are positively correlated because commercial growth raises both.

Therefore in Table 4 we use the change in assessed residential property value as the dependent variable. In addition to the variables used in the previous specifications, we control for the change in the number of housing units between 1980 and 1990. This is never significant in the restricted sample as would be expected since the communities in this sample experienced little new construction.

The first column in Table 4 replicates column (2) in Table 2 except for the change in dependent variable. The positive impact of the increases in the levy is somewhat attenuated using residential property values, but the basic result does not change.

When we use all 334 communities for which data on assessed values by property class are available, preliminary tests establish that we must treat the levy as well as state aid as endogenous. The second column of Table 4 shows the results of this estimation for the maximal set of instruments satisfying the OID test. Despite the fact that we treat the tax levy as endogenous, the effect of the levy is estimated reasonably precisely. The effect of increases in the tax levy is somewhat larger than in Table 2. The coefficient on local receipts becomes more negative and approaches statistical significance.

Table 4
Determinants of % increase in assessed value of residential property (1984–1988)

	Restricted sample[a]	All communities[b]
ΔLevy/100	0.17	0.21
	(0.07)	(0.07)
ΔAid/100	0.49	0.77
	(0.19)	(0.28)
ΔReceipts/100	0.20	−0.20
	(0.09)	(0.10)
F-test on excluded instruments		
Aid	9.00	5.46
Levy	–	18.43
OID test	0.21	3.47
N	178	334

Notes. Two-stage least squares estimates. The explanatory variables are expressed in real per capita terms. All specifications include six community type dummies, six dummies for the timing of the assessments, and change in number of housing units between 1980 and 1990. Standard errors are in parentheses.

[a] Aid endogenous, additional first-stage regressors—constrained in 1982, constrained in 1983.

[b] Aid, levy, endogenous, additional first-stage regressors—constrained in 1982, true tax rate in 1981, and its square, open space ratio in 1984.

6.4. Additional specification checks

In Table 5, we return to using equalized value to provide some additional tests of our specification. The table presents the results using the restricted sample. In all specifications we treat state aid as endogenous and use the two variables measuring constraints as additional first-stage regressors. We present results only from the restricted sample.

The first column addresses the appropriateness of using the percentage change as the dependent variable. Instead, we use nominal equalized value in 1988 and include nominal equalized value in 1984 as an additional explanatory variable. The results are similar to those obtained in Table 2. Increases in all three components of revenue have a positive effect on property values. State aid has the biggest effect followed by increases in the tax levy. However, the coefficient on state aid is poorly determined and thus not statistically significant.

One advantage of the specification in column (1) is that it allows us to think more carefully about the plausibility of the magnitude of the effects. A $1 increase in state aid must be worth at least $1 (ignoring the fact that a small fraction of the $1 comes from income taxes collected in the community). The community could reduce fees by $1 and provide the same public services. If the community is constrained, then it can use the increase in state aid to relax the constraint and receive a benefit valued at more than $1. If the change in state aid is expected to be permanent, then the benefit should be capitalized into the price of the house. If people discount the benefits at 5%, the estimate in column (1) suggests that $1 per year of state aid provides $3.63 per year of benefits. This appears

Table 5
Determinants of equalized value

	1984–1988	1984–1992	1976–1980	1982–1984
	(1)	(2)	(3)	(4)
ΔLevy	62.09	0.10	0.00	0.03
	(14.90)	(0.03)	(0.04)	(0.01)
ΔAid	85.33	0.26	−0.27	−0.14
	(64.52)	(0.18)	(0.11)	(0.11)
ΔReceipts	58.30	0.00	−0.05	0.08
	(21.60)	(0.02)	(0.05)	(0.04)
Equalized val. 1984	2.26	–	–	–
	(0.14)			
ΔLevy 1981–1984	–	–	–	0.02
				(0.01)
ΔAid 1981–1984	–	–	–	0.13
				(0.07)
ΔReceipts 1981–1984	–	–	–	−0.01
				(0.03)
OID test	0.33	6.32	0.09	

Notes. 2SLS estimates (see text for description of instruments). All specifications include six community type dummies and dummies for the timing of assessments. Standard errors are in parentheses. In columns (2)–(4) coefficients are multiplied by 100. (1) Dependent variable—nominal equalized value per capita; (2)–(4) dependent variable—percent change in equalized value per capita.

somewhat high to us but given the imprecision with which the coefficient is estimated is not unreasonable.

One possible explanation for the high coefficients is that people interpreted the change from 1984 to 1988 not as a permanent change in the level but as an indicator of future rates of growth of revenues. In the light of the booming Massachusetts economy, many communities expected state aid and other revenue components to continue growing. To cast light on this explanation, we repeat our analysis for the period 1984–1992. Given the ups and downs experienced by the Massachusetts economy over this period, it is less likely that individuals interpreted changes between 1984 and 1992 as indicators of future changes rather than new levels. By 1992 most communities expected the growth of state aid and other revenues to be low.

In column (2), we present estimates for the percentage change in equalized value from 1984 to 1992 in a manner analogous to those provided in Table 2 for the 1984–1988 period.[3] The coefficients are substantially smaller than those shown in column (2) of Table 2. The impact of state aid is about half of that estimated for the 1984–1988 period and falls short of statistical significance at conventional levels. The impact of increases in the tax levy is about one-third of the estimate for the shorter period but remains highly significant. Both of these results are consistent with the hypothesis that the relation between past growth and expected future growth declined between 1988 and 1992. Finally, we note that the coefficient on receipts falls to zero. Perhaps communities learned how to avoid constraints on fees or improved their ability to set them optimally.

In column (3) we replace equalized values for 1984–1988 (but not the revenue variables) with values for 1976–1980. We then repeat the estimation of Table 2 (column (2)). If our revenue variables were capturing some other factor, then we might expect that they would also explain growth of equalized value between 1976 and 1980. For example, if we were capturing communities on the outer edge of Boston that were growing rapidly, we would expect relatively rapid growth over the entire twelve year period. In fact, the results are quite different from the previous estimates. The coefficients on the tax levy and receipts are effectively zero. While the growth of state aid is associated with changes in equalized value between 1976 and 1980, the sign is the opposite from that obtained for 1984–1988. We therefore conclude that our revenue components are not mere proxies for fast-growing communities.

The fourth column asks whether changes in revenues associated with Proposition $2\frac{1}{2}$ were anticipated and capitalized into housing prices by 1984, thus biasing our results. Despite our concerns about the quality of the equalized value data for the earlier period, we examine changes between 1982 and 1984. We restrict the sample to the 199 communities that reassessed property in either 1981 or 1982 so as to minimize that data quality issues. It is not, however, feasible, for reasons of sample size to further restrict the sample. We redefine the "true tax rate" as the 1981 tax levy/equalized value 1980, to avoid it being correlated with the change in equalized value by construction. We use the true tax rate, its

[3] We use the constraint variables and the true 1981 tax rate variables as first-stage regressors. Preliminary tests rejected the exogeneity of the open space variables and indicated that all four remaining first-stage regressors were required to create a powerful instrument. The F-test on the four excluded regressors is 2.85. The Basmann–Sargan test-statistic is 6.32, well below its critical value at the 0.05 level with three degrees of freedom.

square and its cube as identifying instruments. Because state aid depends on equalized valuations, we must continue to instrument for changes in state aid.[4] We regress the equalized value change on the 1981–1984 and 1984–1988 revenue changes. The results in column (4) show that future tax levy changes were capitalized into housing prices but the effects were small, suggesting that studying the 1984–1988 period does not entail much bias. Moreover, the effect of future state aid changes is large, negative and very imprecisely measured, consistent with the view that the large state aid changes between 1984 and 1988 were not anticipated in 1984.

7. Discussion and conclusion

Our results strongly suggest that communities that were able to increase tax revenues more rapidly in the wake of the Proposition $2\frac{1}{2}$ tax limit in Massachusetts experienced faster growth of property values. Our results are thus consistent with a model in which communities seek to provide the desired level of public services given constraints on their efficiency. Indeed our results for 1984–1992 fit such a model almost perfectly. It is less clear whether constraints on raising revenues through fees were binding. Some specifications show that communities that raised fees more rapidly experienced faster appreciation but others are consistent with the view that there was no relation between fee growth and property value increases as would be expected if communities were unconstrained with regard to setting fees.

Bradbury et al. [2] obtain similar results for the 1990–1994 period using a model that closely resembles ours. Aside from the time period, the papers differ somewhat in their emphasis. They concentrate on the effect of changes in the composition of expenditure while we focus on changes in the structure of revenues. For reasons discussed above, it is impossible to identify separately the impact of changes in revenues and expenditures. Our theoretical model assumes that the distribution of expenditures is efficient given the level and composition of revenues. For purposes of the empirical work, we could have assumed that the distribution of expenditures was independent of the composition of revenues. In contrast, Bradbury et al examine the impact of the distribution of expenditures on housing values. Implicitly, they also assume that the distribution of expenditures is independent of the composition of revenues. Despite this difference in emphasis, our results are similar. Both studies find that increased expenditure/revenue generally raised property values although the impact differs across expenditure types or revenue sources.

The results here and in Bradbury et al. [2] complement Brueckner [4] who found that prior to Proposition $2\frac{1}{2}$ Massachusetts communities showed no systematic tendency to over- or underprovide public goods. His results were consistent with the hypothesis that the level of provision of public goods was efficient and that reductions induced by Proposition $2\frac{1}{2}$ would reduce property values in the more heavily constrained communities.

Our findings underscore the importance of careful thought about the nature of endogeneity when studying Proposition $2\frac{1}{2}$. Vigdor [21] is superficially very close to our

[4] We experimented with also treating the levy change as endogenous. The results were similar but the estimates were more imprecise.

work here and to Bradbury et al. [2]. He regresses various measures of property value increase from 1980 to 1990 on the change in state aid and the tax rate reduction required by Proposition $2\frac{1}{2}$. We can interpret this specification as the reduced-form of a system in which state aid is exogenous and the tax levy and local receipts are endogenous. Yet both the formal link and the statistical evidence suggest that state aid must be treated as endogenous. In contrast the case for changes in the tax levy being endogenous is weaker and that for changes in local receipts quite weak. When we treat state aid as exogenous and local receipts and the tax levy as endogenous, we get results similar to Vigdor's.

We conclude that the evidence is compelling that restrictions on a community's ability to raise taxes did lower its property values. This raises the immediate question as to why Massachusetts voters supported Proposition $2\frac{1}{2}$. In our view the most plausible explanation is that voters simply used the wrong model. Many were convinced that taxes could be cut with only a negligible impact on public services.

There are, however, alternative explanations. One is that voters foresaw that limiting property taxes would shift taxation towards income taxes. For many voters this shift would have been desirable. A second explanation is that the benefits and costs of taxation are distributed unequally. We are unable to investigate whether tax limits help some classes of property such as expensive single-family homes.

Despite this caveat, our results are consistent with models of local public finance that emphasize the importance of the role of local taxes in determining efficient levels of local public goods. They suggest that tax limits may have severe adverse effects that are not anticipated by their proponents.

Acknowledgments

This research was supported in part by NSF grant SBR-9515052. We are grateful to Eli Berman, Harvey Beth, Kathy Bradbury, Jan Brueckner, John Conley, Dennis Epple, Ed Glaeser, Hideo Konishi, Jim Poterba, Brian Sullivan, two anonymous referees and participants in the Boston University Applied Microeconomics workshop and the 1997 ASSA meetings for helpful comments and discussions. We thank Roger Hatch, John Sanguinet, and Julie Slavet of the Massachusetts Department of Revenue and Donald Buckhotz and Scott Keese of the Massachusetts Taxpayers Foundation for providing data and other assistance. The usual caveat applies.

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